

Mock exam

Algebra · 60 minutes · formula sheet and calculator allowed

Three questions

Instructions

- Attempt all three questions. Show all working.
- You may use the formula sheet and a calculator.
- Time: 60 minutes. Start the clock when you turn the page.
- Each question is graded N0–E8; the total is out of 24.

After the exam

- Mark each question with the assessment schedule: count the u, r and t evidence, then give it a grade from N0 to E8.
- Add the three grades. Cut scores move a little each year: Achievement from about 7–8, Merit 13–14, Excellence 19–20.
- For every mark lost, write the L-number from your handouts next to it.

QUESTION ONE

- (a) Simplify each expression, leaving your answer with positive indices.

(i) $\sqrt[5]{\frac{4(2n)^3}{n^8}}$

(ii) $\left(\frac{n^3}{16n^6}\right)^{-0.5}$

- (b) $\frac{20x^2 - 22x + 6}{5x^2 - 3x}$ can be written in the form $A + \frac{B}{x}$, where A and B are integers.

Find the values of A and B .

- (c) The value, V , in dollars (\$), of a laptop can be modelled by $V = 40 + ke^{-0.5t}$, $t \geq 0$, where t is the time in years since the laptop was purchased. The original price of the laptop was \$900.

How long does it take for the laptop's value to be reduced to 50% of the original value?

- (d) $2y = 2x + 29$ is a tangent to the quadratic $x^2 - 2ky + 32k = 0$, where k is a non-zero constant.

Find the value of k and determine where the quadratic crosses the y -axis.

QUESTION TWO

- (a) (i) Find m if $\log_2(3m + 1) = 4$.

- (ii) Solve the following equation:

$$3 \log_x(64) = 6.$$

- (b) Find an expression for p in terms of x if $\frac{5^{7x+6}}{25^{-x}} = 125^p$.

- (c) Find the value of $6 + \log_b\left(\frac{1}{b^3}\right) + \log_b(\sqrt{b})$.

(d) Using an algebraic method, solve $4^x - \frac{10}{4^x} = 3$.

QUESTION THREE

- (a) (i) Solve the following equation:

$$5x^2 + 13x = 6$$

- (ii) Simplify:

$$\frac{3x^2 - 10x + 8}{9x^2 - 16}$$

- (b) Find the range of values of p for which the graph $f(x) = 2x^2 + 8x + p$ does not cross the x -axis.

- (c) $\frac{x^2 + 2x + k}{x^2 + 7x + 10} = \frac{x - 3}{x + 2}$ where k is a constant.

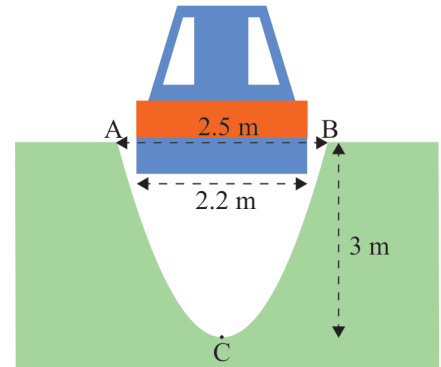
Work out the value of k .

- (d) The diagram below shows the cross-section of a canal and narrowboat floating in the canal.

- The surface of the water (between points A and B) measures 2.5 m across.
- The canal is 3 m deep, at the deepest point C.
- The cross-section of the canal can be modelled as a quadratic curve ACB.
- The cross-section of a narrowboat on the canal can be modelled as a rectangle with a width of 2.2 m.
- The narrowboat must maintain a constant depth below the water of 1 m in order to float.



Source: www.pxfuel.com/en/free-photo-xerdj



Will the narrowboat be able to float in this canal?

- (e) An open box (i.e. with a base but no lid) has been designed to tightly fit a cylindrical candle. The surface area of the five surfaces of the box is equal to the total surface area of the candle.

Write an expression for the height, h , in terms of the radius, r , and π .

(Surface area of a cylinder = $2\pi r^2 + 2\pi rh$.)

